MedFi

Decentralized Health Savings & Micro-Loan Protocol on Botchain

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1. Executive Summary

MedFi is a decentralized health savings and micro-loan protocol deployed on Botchain. It allows individuals — particularly in underserved African communities — to save small amounts of BOT (Botchain's native token), earn MedFi Tokens (MFT) as on-chain health reputation, and borrow low-cost health loans backed by a community pool and a personal guarantor system.

MedFi is inspired by the community health savings groups (esusu, tontine, chama) that have existed across Africa for generations. It brings those time-tested models on-chain, making them transparent, tamper-proof, and accessible to anyone with a smartphone and a Botchain wallet.

2. Problem Statement

2.1 Healthcare is Unaffordable Without Credit

In sub-Saharan Africa, over 600 million people have no health insurance. When illness strikes, families must pay out-of-pocket — sometimes selling assets or borrowing from informal lenders at interest rates of 50-200%. This drives healthcare avoidance and medical debt cycles.

2.2 No Access to Formal Credit

Rural and peri-urban residents are routinely excluded from bank credit. Without a credit history or collateral, formal micro-loans are out of reach — even for a 10 USD doctor's visit.

2.3 Fragile Informal Systems

Existing community savings groups are informal, trust-based, and vulnerable to mismanagement or collapse. There is no transparent record, no smart enforcement, and no composability with healthcare providers.

3. MedFi Solution Architecture

3.1 Core Primitives

Primitive	Description	On-Chain Data
Member Profile	Registered wallet with savings history	name, totalSaved, streak, MFT, activeLoans
Savings Pool	Shared BOT pool earning from loan fees	totalPoolBalance, per-member deposits
Loan	Health micro-loan with guarantor security	amount, guarantor, stake, dueDate, status
Provider Registry	Verified clinic/pharmacy addresses	name, category, active flag
MFT Token	Health reputation earned on-chain	balance per member, mint rules

3.2 Savings Mechanic

Members deposit BOT into the MedFi pool at any time. Each deposit earns 10 MFT. A weekly savings streak (at least one deposit per ISO week) grants a 5 MFT bonus after 4 consecutive weeks. A member's total saved determines their maximum loan: up to 3x their cumulative savings.

3.3 Loan Lifecycle

- Step 1 — Borrower calls requestLoan(amount, guarantorAddress, durationDays)
- Step 2 — Guarantor calls stakeAsGuarantor(loanId), locking half the loan amount in BOT
- Step 3 — Contract disburses full loan amount to the borrower from the pool
- Step 4 — Borrower repays via repayLoan(loanId) within the due date
- Step 5 — 2% flat fee is retained in the pool; guarantor's stake is returned
- On-time repayment earns 20 MFT; late repayment earns 10 MFT
- Default (past due date): owner calls markDefault(); guarantor stake covers partial loss

3.4 Direct Provider Payments

Members can call payProvider(providerAddress) with a BOT amount to pay any verified clinic or pharmacy directly. The payment is recorded on-chain, creating an auditable health spending history tied to the member's wallet.

3.5 MFT – MedFi Token

Action	MFT Earned	Notes
Deposit (any amount)	10 MFT	Every successful deposit call
Streak bonus (4+ weeks)	+5 MFT	Added on top of deposit reward
On-time loan repayment	20 MFT	Repaid before dueDate

Late repayment	10 MFT	Repaid after dueDate but before default
Default	0 MFT	Guarantor stake partially liquidated

MFT is a non-transferable (soulbound) on-chain reputation score in v1. In v2, MFT holders will be able to redeem tokens for discounts at partner providers and participate in DAO governance votes.

4. Milestones & Activities on Botchain

Milestone	Key Activities	Timeline	Success KPI
M1 Testnet Deploy	Deploy MedFi.sol on Botchain testnet Write and run full unit test suite Set up multisig owner wallet	Month 1	Contract verified on Botchain explorer 100% test coverage
M2 Frontend Launch	Launch index.html dApp Wallet connect integration Deposit and loan request flows live	Month 2	Live dApp accessible on Botchain 50 test wallet interactions
M3 Pilot Members	Onboard 50 pilot members in Lagos & Nairobi Deposit campaign: min 0.5 BOT/week First 10 loans disbursed	Month 2-3	50 registered members 10 active loans >0 BOT pool balance
M4 Provider Network	Partner with 5 clinics and 3 pharmacies Add provider wallets to registry Test direct provider payments	Month 3-4	8 verified providers First on-chain provider payments
M5 Repayment Cycle	First loan cohort reaches due date Measure repayment rate Issue MFT for on-time repayments	Month 4	>80% repayment rate MFT distributed to 40+ members
M6 Mainnet Launch	Deploy to Botchain mainnet Marketing to health NGOs and co-ops Target 300 members and 100 BOT TVL	Month 5-6	300 members 100 BOT TVL 12+ provider partners
M7 DAO Governance	Introduce MFT-weighted voting Community votes on loan parameters Onboard community assessors for defaults	Month 7-9	First DAO vote passed 5 community assessors

5. Economic Model

MedFi charges a simple 2% flat fee on each loan repayment. This fee stays in the pool and is distributed to depositors over time, creating a yield for savers. The protocol takes no protocol-level cut — 100% of fees benefit the community pool.

Flow	Amount	Destination
Loan fee (per repayment)	2% of loan amount	Community pool
Guarantor stake (no default)	50% of loan, returned	Back to guarantor
Guarantor stake (default)	50% of loan, kept	Pool (covers loss)
MFT minting	Earned, not bought	Member reputation wallet

6. Risk Management

6.1 Guarantor Collateral

Every loan requires a guarantor to lock 50% of the loan amount on-chain before disbursement. This social-collateral model mirrors traditional African savings groups, where community accountability drives repayment. A guarantor who approves a bad loan risks losing their stake.

6.2 Savings-Linked Borrowing

The 3x savings cap prevents overleveraging. A member with 1 BOT saved can borrow at most 3 BOT. This ties borrowing capacity to demonstrated saving discipline, reducing default risk.

6.3 Loan Limits

- Maximum 1 active loan per member at a time
- Maximum duration: 180 days (6 months)
- Minimum duration: 7 days
- Maximum loan: 3x cumulative savings

7. Technical Stack

Layer	Technology	Purpose
Smart Contract	Solidity ^0.8.20 on Botchain	Savings, loans, MFT, provider registry
Frontend	HTML5 / CSS3 / Vanilla JS	Member-facing dApp
Wallet	Botchain-compatible wallet	Transaction signing and identity
Evidence (v2)	IPFS CID on-chain	Medical receipts / provider invoices
Governance (v2)	MFT-weighted DAO	Parameter changes, provider votes

8. Conclusion

MedFi is a simple, honest protocol: save a little, borrow when sick, repay and build your reputation. It requires no bank account, no credit bureau, and no middleman — just a Botchain wallet and a willing community member to stand as your guarantor.

By anchoring health savings and loans on Botchain, MedFi creates the financial infrastructure that millions of Africans need but have never had access to. Every deposit, loan, and repayment is permanent, public, and composable — a foundation for a fairer healthcare economy.